

Forces and
energy

Types of energy:

- Nuclear energy
- Thermal / Heat / Internal
- Electrical
- Light
- Sound
- Chemical
- Kinetic
- Gravitational potential
- Elastic potential

* Energy can be transferred:

- Mechanically, by a force moving something
- Electrically by a current
- By heating because of a temperature difference
- By radiation such as light waves and sound waves.

Conservation of energy: Energy cannot be created or destroyed, it can only be transferred from one type to another.

Gravitational potential energy: mass $\times$ Gravitational field strength $\times$ height

$$\underset{\substack{\text{J} \\ \downarrow}}{\text{GPE}} = \underset{\substack{\text{kg} \\ \downarrow}}{m} \underset{\substack{\text{N/kg} \\ \downarrow}}{g} h \rightarrow m$$

Kinetic potential energy: $\frac{1}{2} \times \text{mass} \times \text{velocity}^2$

$$\underset{\substack{\text{J} \\ \downarrow}}{\text{KE}} = \frac{1}{2} \underset{\substack{\text{kg} \\ \downarrow}}{m} v^2 \rightarrow (\text{m/s})^2$$

Efficiency: A measure of how useful a device converted energy

$$\text{Efficiency} = \frac{\text{Output energy}}{\text{Input energy}} = \frac{\text{Useful}}{\text{Output}}$$

* Efficiency is unitless

Work = Force $\times$ distance

$$\underset{\substack{\text{J} \\ \downarrow}}{W} = \underset{\substack{\text{N} \\ \downarrow}}{F} d \rightarrow m$$

* Force and distance must be parallel

Work = Energy

$$F d = m g h$$

weight

Power: Rate of doing work / Rate of transferring energy.

$$\text{Power} = \frac{\overset{\substack{\text{J} \\ \downarrow}}{\text{Work}}}{\underset{\substack{\text{s} \\ \downarrow}}{\text{Time}}} = \frac{\text{Energy}}{\text{Time}}$$

Watt
W

$$\text{Efficiency} = \frac{\text{Useful power output}}{\text{Total power input}}$$

Useful energy may be obtained, or electric power can be obtained from:

1- Chemical energy stored in fossil fuels:

Disadvantages: * Produce waste gases that pollute the atmosphere
* Non-renewable
* Cause global warming

2- Nuclear fuels: Energy is released by fission reaction: a process in which the nuclei of uranium atoms are split.

Disadvantages: * The waste from nuclear fuel is very dangerous and remains radioactive
* Power stations are very expensive to build and to decommission
* Non-renewable

3- Hydroelectric energy: A river falls behind a dam. Water flowing down from the lake turns the generator

Disadvantages: * Expensive to build
* Few areas in the world are suitable
* Flooding land & building dams causes environmental damage.

4- Tidal energy: A lake fills when the tide comes in and empties when it goes out.

Disadvantages: Same as hydroelectric

5- Wave energy: Generators are driven by the up-and-down motion of the waves at sea.

Disadvantages: * Difficult to build
* Few devices have been successful

6- Wind energy: Generators are driven by wind turbines

Disadvantages: * Large, remote, windy sites are required.

* Winds are variable

* The wind turbines are noisy and can spoil the landscape.

7- Geothermal: Water is pumped down to hot rocks deep underground and rise as steam.

Disadvantages: * Deep drilling and expensive

8- Solar energy: Solar panels absorb the energy radiated from the Sun and use it to heat water. Solar cells are made of materials that can deliver an electric current when they absorb the energy in light.

Disadvantages: * Variable amounts of sunlight in some countries

* Solar cells are expensive

* Solar cells must be large to deliver useful amounts of power.

9- Biofuel: Fuels made from plant or animal matter.

Disadvantages: * Huge areas of land are needed to grow plants

* Radiation from the sun is the main source of energy for all energy resources except for:

1- Geothermal

2- Nuclear

3- Tidal

* Energy in the sun is released by nuclear fusion.